

Seanie Stack

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Professional Summary

Final Year BSc Computer Systems student with 8 months of professional software development experience with a focus on backend and systems engineering. Designed and built a full programming language (lexer, parser, tree-walk interpreter, embedding API) in Java with ANTLR4, and shipped production code in Java / Spring Boot during an internship where a single performance fix reduced all API response times by 91 %. Comfortable across Java, Python, C, Rust and Go, with hands-on experience with Linux, Docker, MySQL, PostgreSQL, RabbitMQ and Redis. Eligible to work in Ireland and the UK.

Education

University of Limerick

Sept 2022 – Jan 2027

BSc Computer Systems 3.11 QCA Second Class Honours

Professional Experience

Ostia Solutions

Bray, Co. Wicklow

Software Development Intern

May 2024 — Jan 2025

- Worked across test development and performance testing during an 8 month placement, contributing to the quality and reliability of production Spring Boot Java services.
- Designed and implemented automated test suites using JUnit, JMeter and Postman, increasing test coverage and catching bugs earlier in the release cycle.
- Identified and fixed a cache misconfiguration across all major API endpoints, reducing average response time by 91 % and improving throughput under load.
- Investigated and reported performance bottlenecks using load testing and profiling, while working alongside senior developers to ship fixes.
- Operated in an agile team environment using git for version control and standard CI workflows for build, test and deployment.

Projects

RJScript — Final Year Project (Java, ANTLR4)

2025 - 2026

Github

- Designed and implemented a dynamically-typed scripting language from scratch in Java (12,000 lines of code), including lexer, parser, AST, and tree-walk interpreter built on ANTLR4.
- Implemented core language features including variables, arithmetic and logical operators with short-circuit evaluation, functions with closures, lexical scoping, strings, floats, arrays with negative indexing, and modulo.

- Built a two-tier standard library and module system, plus a Java embedding API (RJScripEngine) allowing host applications to execute RJScrip code and exchange values with the JVM.
- Added source-location-aware error reporting and configurable debug flags to improve the developer experience when writing and debugging RJScrip programs.

E-Library Microservice Backend (Java, Spring Boot, RabbitMQ, Docker)
Github

2026

- Built a microservices-based e-library application split into a core service (catalogue, loans, users) and a support service (fines, notifications), communicating asynchronously via RabbitMQ.
- Used RabbitMQ message queues to decouple fine processing and user notifications from the main request path, improving responsiveness and resilience to downstream failures.
- Containerised all services with Docker and orchestrated local deployment with Docker Compose, including dependency services such as the message broker and database.

Technical Skills

Languages: Java, Python, C, Rust (basic), Go (basic)

Frameworks & libraries: Spring Boot, ANTLR4, JUnit, Mockito, JMeter

Tools & Platforms: Git, Docker, Docker Compose, Maven, Gradle, RabbitMQ, Linux (Arch)

Databases: MySQL, Redis, PostgreSQL

Concepts: Compilers and interpreters, language design, backend systems, microservices, performance testing, automated testing, message queues, REST APIs